

Chapter 10

Caste, Colonialism and Schooling: Education in British India

Latika Chaudhary*

May 2012

1 Introduction

The subject of education and human capital is among the most neglected in Indian economic history. Few economics articles have been published on the topic and even the extremely detailed Cambridge Economic History of India does not offer a chapter on education. There are historical studies on colonial education, but these are few in number and qualitative in nature. For example, Nurullah and Naik (1951) offer a dated but excellent overview of the major official recommendations and their impact on education development in British India. They emphasize how specific policies, such as the inadequate support of indigenous education, among other factors constrained the expansion of mass education. Scholars like Bellenoit (2007) have studied the effect of missionary education on Indians highlighting the complicated nature of the relationship. Basu (1974) has explored how the increase in secondary education and the related rise in political nationalism influenced subsequent state policies. Much has also been written on how English education influenced urban Indians (for example, B.T. McCully 1960). Across these studies, the general consensus holds Imperial policy responsible for the inadequate progress. But, as argued by Whitehead (2005) the historical literature is much in need of “plodding” in archives to substantiate, refine or refute the claims of India’s educational historians.”

Apart from echoing the findings of the historical literature, Indian economic history has largely been silent on the subject. Part of the neglect dates to an older era when

* Assistant Professor of Economics, Scripps College, 1030 Columbia Avenue, #4072, Claremont, CA 91711.
Email: latika.chaudhary@scrippscollege.edu

economists disproportionately studied physical investments and the concept of education as a human capital investment was new. Nonetheless, recent decades have clearly established the importance of education to economic growth and development (Schultz 1962 and 1971, Becker 1964, Mincer 1974, Easterlin 1981, and Benhabib and Spiegel 1994 among many others). An educated labor force has been linked to greater worker productivity, a faster ability to adopt new technologies and lower crime. Even in traditional agricultural economies, education has been linked to development because literate farmers may be quicker to adopt superior inputs and engage in greater information sharing in turn leading to higher productivity (Schultz 1983). Understanding the economics of India's schooling record is important in and of itself, and also because it can inform debates surrounding the Indian economy under the Raj.

This article briefly reviews the institutional history, presents data on enrollment and literacy, and discusses how colonial policy interacted with local conditions to influence the trajectory of Indian education. Two caveats are in order. First, similar to other studies the focus is on British India on account of data and space constraints. Education in the Princely States is an extremely under researched area of study.¹ I hope future work will address the many complicated issues surrounding the provision of schooling in these States. Second, this article is neither a review of the historical literature nor of the underlying data sources. Rather, the goal is to use quantitative data and examine specific factors that hurt the expansion of mass education in the late 19th and 20th century.

While this period begins with British officials constructing the necessary policies, it ends with elected Indian representatives drafting education policy. After the East India Company handed control of British India to the Crown in 1858, the Government of India (GOI) under the authority of the India Office in London was the relevant decision-making unit managing the provision of education among other public services. Beginning in the 1870's, the GOI decentralized public services including education to provincial governments followed by another wave of decentralization to district and municipal boards in the 1880's. While the GOI and provincial governments were represented by British administrators, Indians served on the boards in an elected and nominated capacity. The Montague-Chelmsford reforms of 1919 ushered in a limited form of self-government. As part of the reforms, education came under the purview of provincial legislative councils composed of elected Indian representatives. A more comprehensive form of self government was introduced in 1935 under Provincial Autonomy. Despite these political changes, enrollment and literacy rates remained stable and disappointingly low for most of the colonial period.

¹Iyer (2010) is among the first to offer a serious quantitative assessment of the effects of indirect rule on economic outcomes including literacy in the Princely States.

2 Colonial Policies

Over the 19th century, the indigenous system of schooling in British India was replaced by the new state system of education developed initially by the East India Company, controlled by the British Crown from 1858 to 1919 and taken over by elected Indian ministers after the 1919 Montague-Chelmsford reforms. Under the former indigenous system, schools were of two types—elite religious schools for students interested in a lifetime of higher learning and local elementary schools where village boys were introduced to the 3 R’s in the vernacular medium.² Due to lack of official patronage, both the elite schools and local indigenous schools declined over this period. Some historians suggest 8 to 12% of the male population was literate, but I interpret these estimates with caution because a systematic enumeration of literacy did not begin until the early 20th century.

Wood’s Education Despatch of 1854 was the first official document akin to a national education policy, which outlined the Company’s role in the provision of schooling.³ The Despatch created provincial education departments and established guidelines for the development of schools at the primary, secondary and collegiate level. While earlier policies had promoted “a very high degree of education for a small number of natives” in the English medium,⁴ the Company now emphasized the importance of expanding vernacular primary education for the rural masses.⁵ Given the high costs of building such a system, the Despatch introduced public subsidies known as ‘grant-in-aids’ to partially support privately managed schools that came to be known as aided schools. Under this system, grants were available to schools that followed a secular curriculum (religious neutrality), were under private management, and open to public inspection. Grants could be allocated to specific charges, for example teacher salaries or buildings. However, they could not cover all the operating expenditures of the institution. The Despatch also mandated that eligible schools charge fees, nominal if necessary, to encourage regular attendance. Provincial governments were given substantial leeway in framing grant eligibility rules. By encouraging grant-in-aids, the East India Company created an important role for private participation in the education sector.

²For details see Adam (1838), Nurullah and Naik (1951) and Basu (1982).

³The East India Company was generally indifferent to the provision of schooling until the early 19th century when they set aside public funds for centers of classical learning and the promotion of western sciences in the English medium. Although no vigorous efforts were made to encourage local indigenous schools, the Company promoted English medium schools with the expectation that they would produce an elite group of educated Indians who could work in the colonial administrative offices (Nurullah and Naik 1951).

⁴See Despatch to Government of India on the Subject of General Education in India (1854).

⁵English was established as the official language of instruction in 1835 by Governor General Bentinck following Macaulay’s infamous minute that strongly criticized Oriental languages and literature, while promoting the study of western sciences and philosophy in the English medium. See Ghosh (2000) for details surrounding debates between the Anglicists and Orientalists.

After the Revolt of 1857, the Government of India under the Crown accepted the provisions of the 1854 Despatch.

Beginning in the 1860's, a new state system of education emerged whereby publicly financed and managed schools (provincial government and local board schools) functioned alongside privately managed aided and unaided schools. Provincial governments directly controlled their schools, the vast majority of which were secondary schools and colleges. Primary education was decentralized to rural district and urban municipal boards in the early 1880's. The boards managed their public schools and also authorized grants to private aided schools. Private unaided schools did not receive any public money, but they were classified under the new state system because they were open to inspection and their students were allowed to take public examinations. Many of the former indigenous schools disappeared over this period, some were successfully converted into aided schools and the rest were classified as private unrecognized schools because they did not conform to official standards and were often religious in nature.⁶

Since local boards were financed through public revenues, colonial policy largely determined the placement and development of provincial government and board schools. That said, upper castes and landed elites were unequally represented on local councils and in principle they could influence public allocations to an extent.⁷ In comparison, the British had limited control over aided schools because private individuals pooled the necessary resources, set up the school, applied for a public grant and then managed the school. Both aided and unaided schools were privately managed either by Indians or by missionaries. Although missionaries controlled a large number of private schools in the early 19th century, their share declined over the second half of the century as private Indian participation increased.⁸

There was a dramatic increase in the number of private English secondary schools and colleges in the late 19th century fueled by strong private demand among Indian elites hoping to secure lucrative government jobs after graduation.⁹ British officials recognized the

⁶From 1855 to 1882, the number of private indigenous schools declined by almost 50% from 49,524 to 25,166, while government and aided schools increased to 89,005 by 1882. See the Hunter Indian Education Commission Report (1883). By 1917, less than 10% of pupils were enrolled in private indigenous schools. Since private indigenous schools were uninspected, and frequently opening and closing, their data are not considered accurate.

⁷Since British officials were often board chairmen, they had the authority to either override or defer to non-official opinions.

⁸After the Despatch, missionary efforts were not treated favorably by colonial officials. Moreover, they had to compete with newly established government schools and Indians (especially the upper castes) preferred the secular nature of government schools. In 1883, the government explicitly stated their preferences for promoting Indians in the field of private education relative to missionaries and consequently missionaries reduced their efforts to the maintenance of a few schools.

⁹For example, the number of English secondary schools and arts colleges more than doubled from 1881/82 to 1921/22 from 2,133 to 4,904, while the number of pupils more than quintupled from 149,233 to 823,416

growth of secondary education was far outpacing primary education, and subsequent policy reports highlighted the need to expand mass primary schooling.¹⁰ Local boards were encouraged to build new schools or offer more public grants to aided schools. Various schemes were introduced to increase schooling among groups with below average literacy such as Muslims, tribal groups, lower castes and women.¹¹ In addition, larger public revenues were targeted toward primary schooling in the early 20th century along with a new focus on quality improvements and more state control.¹² These positive measures notwithstanding, British officials largely resisted the idea of free primary education throughout this period and delayed the introduction of compulsory schooling laws.¹³ Beginning in 1918, some form of compulsory schooling was gradually introduced just as education was transferred to the control of provincial governments under elected Indian ministers. These changes introduced by the 1919 Montague-Chelmsford reforms mark the end of direct colonial responsibility for education.

Under Dyarchy, provincial governments dramatically increased public spending on education. By 1931-32, public expenditures at Rs. 168,419,016 were over three-times their 1911-12 level. Since price levels declined over these decades, this increase was significant in both real and nominal terms. With these additional funds, numerous primary schools were constructed in the 1920's. Many villages now had at least one primary school if not multiple. Unfortunately, this expansion of schools did not translate into similar gains in literacy: 1931 literacy was only marginally above that in 1921. In reviewing the progress of education, the Hartog Committee (1929) attributed the discrepancy to the rapid expansion of poor quality, single teacher schools in many parts of the country. Consequently, there was a consolidation of schools in the 1930's and some of the poor quality schools were shut down. While the subject of education enjoyed popular and political support in the 1920's and 30's, it gave way to other pressing concerns after Provincial Autonomy was introduced in 1937.

(Basu 1974, p. 105).

¹⁰See Education Commission Report (1883) and Progress of Education in India: Quinquennial Reviews. For example, the Indian Education Commission Report of 1882 made primary education a subject of critical importance with a declaration that "elementary education of the masses, its provision, extension, and improvements, to be that part of the educational system to which the strenuous efforts of the State should now be directed in a still larger measure than heretofore."

¹¹Larger public grants were made available to schools in "backward districts", scholarships were introduced to encourage schooling among these groups, and training schools for teachers were established.

¹²In the early 1900's, colonial officers began to view secondary schools and colleges as a breeding ground for political unrest against colonial rule and for rising nationalism among educated Indians. Consequently, numerous acts were passed both to increase efficiency and control over public institutions.

¹³Gokhale, an Indian champion of primary education, introduced a private bill in 1911 that outlined a modest system of compulsory education for boys between the ages of six and ten. But the bill was rejected on the grounds that there was no popular demand for such a measure.

3 Enrollment and Literacy

The numerous commissions, official acts and policy recommendations of the colonial period translated into a mixed picture of lopsided progress. While the goal of extending basic education to the masses was largely unrealized, huge gains of perhaps uncertain quality were made at the secondary and post-secondary level.

Figure 1 graphs total enrollment as a percentage of the population from 1850 to 1940 illustrating a picture of steady progress. That said, the increase is not particularly significant because India moved from an extremely low level of 0.014% in 1853 to 4% in 1940, which was still a low level. By pooling students across primary, secondary and post-secondary institutions, the aggregate enrollment also misses differences between educational levels. Table 1 reports enrollment rates in recognized institutions by level and province. While official reports and data sources such as the Statistical Abstracts document the total number of students in the mid-19th century, a systematic enumeration of enrolled students began after the Hunter Education Commission (1883) noted the importance of collecting such statistics. Following the report, the first Quinquennial Report on the Progress of Education in India was released in 1886-87 and subsequent Quinquennial Reports were published up to 1947. The series in table 1 use these data from 1891 to 1941. The Quinquennial Reports do not report the age break down of the population, but suggest the primary school age population (ages 5 to 10) accounted for 15% of the total population. I use this figure to construct the school age population in each province. Table 1 only counts students enrolled in recognized institutions. Students enrolled in unrecognized indigenous schools such as *maktabs* and *tols* are excluded.

The aggregate enrollment patterns provide strong evidence of India's limited achievement at the primary level, but relatively superior performance at the secondary level. As late as 1891 only one out of ten primary school-age children were enrolled in any type of school. The number of students enrolled steadily increased in the 20th century, but even by 1941 only one-third of school age children (35%) were enrolled in school with sharp regional differences. While primary school enrollment clearly increased in this period, the gains at the secondary and collegiate level were more remarkable. Secondary school enrollment more than quadrupled between 1891 and 1941 with more than 6 percent of school-age children attending secondary school by 1941. Secondary schools include middle schools and high schools, both of which had attached primary classes offering higher quality instruction compared to vernacular primary schools.¹⁴ The gains at the post-secondary level were even more

¹⁴On average, less than 10% of total primary level pupils were enrolled in primary departments of secondary schools. For example, 8.9% and 9.5% of total primary level pupils were in secondary schools in 1907 and 1912 respectively. See Progress of Education in India: Quinquennial Review (1907-12).

dramatic: college enrollment increased more than seven-fold with most of the gains accruing to arts colleges. These statistical patterns match qualitative accounts of urban Indians flocking to post-secondary institutions in large numbers.

The enrollment rates are especially striking when placed in an international context. Using information in Lindert (2004), table 2 documents primary and secondary enrollment rates in India compared to more advanced economies (for e.g., France and the United States) and countries at comparable levels of development (for e.g., Mexico and Brazil). Lindert's series for British India exclude private unaided schools. He also defines the school-age population as ages 5 to 14 across countries. The 15% school-age population is useful in the Indian context to evaluate the number of primary school-age children attending school, but to generate comparable statistics table 2 reports enrollment rates from table 1 adjusted for a school-age population of ages 5 through 14.¹⁵ Table 2 reports data from the Quinquennial Reviews and Lindert's series for British India. Since Lindert excludes private unaided schools, his series underestimate Indian enrollment. Despite their differences, both series highlight India's poor performance at the primary level. British India had one of the lowest primary school enrollment rates in every decade averaging 9% and far below the 80% enrollment in advanced parts of Europe. Among the set of less developed economies, Indian primary school enrollment (14% in 1930) is lower than Brazil (22%), Japan (61%), Mexico (37%) and Chile (56%). In comparison, secondary school enrollment in British India is comparable to France, and even exceeding the UK in the 1890's and 1900's. In 1910, the percentage of the school-age population enrolled in secondary schools was 1.4% for India, 1.4% for France, and 2.1% for the UK. These differences persist even after accounting for primary students in Indian secondary schools. While secondary school enrollment in British India was comparable to developed countries, primary education was far below average in the early 20th century. More research is needed to identify the effects of such lopsided progress on economic outcomes.

The aggregate enrollment levels mask the tremendous regional heterogeneity within India. At every level, the more advanced coastal provinces of Bengal, Bombay and Madras outperformed the interior provinces of Bihar and United Provinces. Differential returns and the related private demand for primary education clearly influenced these regional patterns as did colonial policies related to land revenue (Chaudhary 2010). Primary school enrollment in Bombay and Madras averaged 15% (adjusted for population ages 5 to 14), and was higher than Brazil (12%), but still below Mexico (22%), Chile (33%) and Japan (50%). Even the

¹⁵The population data in the Quinquennial Reviews are not reported by age and include data on some Princely States in their counts. Hence, they are not exactly comparable to the census that reports population by age. To get around this issue, I used the 1911 census to calculate the proportion of children ages 5 to 14. On average, the 5 to 14 age group accounted for 25% of the population. So, I used 25% of the total population in each province to generate the enrollment rates in table 2.

best performers in British India were unable to match the average educational attainment in the more progress developing economies of this era.

An examination of Indian literacy paints an equal, if not more, dismal picture. The census began collecting literacy information in the late 19th century, but these early censuses classified the population into individuals learning at a school, college or home; individuals who were literate but not under instruction; and individuals who were illiterate and not under instruction. Subsequent censuses noted several problems with this classification scheme. Hence, in 1901 individuals were categorized as literate or not, but there was no consistent standard and local officials used their own standards to determine literacy. Beginning in 1911, a uniform standard was adopted—an individual who could read and write a short letter was enumerated as literate. Census officials note the definition was closely followed across provinces and in subsequent censuses. Due to the changing classifications, I focus on the post-1901 literacy counts in the ensuing discussion.

Table 3 reports literacy by gender, province and year in the top panel. Male literacy increased by 5 percentage points between 1901 and 1931, but averaged only 17.8% in 1931. Female literacy was incredibly low at 3% in 1931 increasing 2 percentage points from 1% in 1901. Even the limited gains in primary school enrollment in the 20th century did not translate into significant literacy gains. While primary school enrollment almost doubled between 1901 and 1931, overall literacy merely increased from 13% to 18% for males, and 1% to 3% for females. Colonial officials were well aware of this discrepancy and numerous reports bemoaned the tremendous waste in the primary education sector. The main problem was insufficient attendance. For example, out of 100 students beginning primary school in 1922-23, only 18 were in class four in 1925-26. The remaining had either dropped out or in a few cases stagnated at a lower grade. If we assume it takes at least four years of regular attendance from class one to class four to acquire functional literacy, only 18% of primary school students were becoming literate. The wastage was even worse in female schools (Hartog Committee 1929) where girls were regularly pulled out of school for early marriage among other reasons. Although the significant and persistent divide between male-female educational attainment was recognized historically, no effective solutions to the problem emerged in the colonial period.

Although average literacy was low, the small number of literates were highly educated as evidenced by the rates of English literacy. Thirteen percent of male literates could read and write in English, and 10 percent of female literates also possessed English literacy. Instruction in the English medium was more common at the secondary level and these patterns further reinforce India's high level of secondary education. Since wastage was less common here, the secondary system was also more successful at increasing literacy.

The bottom panel of table 3 reports literacy by religion and caste in 1931, which highlight the tremendous variation across social groups. Certain religions such as Christians and Jains were among the most literate in colonial India. Jains were predominately an affluent trading class and education was almost an economic necessity. At the other end of the spectrum, tribal groups living in geographically remote parts of the country had the lowest literacy under 1%. Average Muslim literacy at 6.4% was below Hindu literacy at 8.4%, but there were significant regional differences. Muslim literacy was negatively correlated with the presence of other Muslims both across provinces and across districts within provinces (Chaudhary and Rubin, 2011). For example, Muslims in Madras had higher literacy than those in Punjab or United Provinces, provinces with larger Muslim populations.

Among Hindus, there were large differences by caste. Brahmans at the upper end of the caste spectrum averaged 33%, while depressed castes averaged 1.6%. Despite the regional differences, the inter-caste patterns are visible across provinces. In 1931 80% of Brahman males could read and write in Bombay and Madras, while lower caste literacy ranged from a high of 5% in Bengal to a low of 0.5% in United Provinces. Brahmans were not the only upper caste with above average literacy. They were the most literate Hindu caste in Bombay, Central Provinces and Madras, but they were substantially below the Kayasthas in UP (45% compared to 16%) and Baidyas in Bengal (63% compared to 42%). That said, these differences among the educated castes are small compared to the differences between the upper castes and the depressed castes. The inter-caste differences should be viewed as a function of economic, political and social differences between the castes. The depressed castes for example were poorer on average and until the 1920s also had limited political voice. Thus, it is difficult to separate the effect of caste from the economic and political differences between the upper and lower castes.

This overview highlights four basic features of India's educational development. First, the limited achievement at the primary level. Second, the relatively high achievement at the secondary level. Third, the large differences in literacy by caste, gender and religion. Fourth, the regional differences within British India. The next section proposes explanations to account for these patterns, but more research needs to be done before we can draw definitive statements.

4 Returns, Colonial Policies or Caste?

4.1 Returns to Schooling

In standard economic models, rational individuals invest in schooling when returns (i.e., wages) exceed the direct costs of school fees and the opportunity costs in terms of foregone

wages. Applying this model to India suggests low returns to primary education are perhaps one explanation for the low level of literacy. Colonial India's large agricultural sector and relatively low GDP per-capita probably weakened the demand for basic education. Moreover, a heavy reliance on child labor during the agricultural season increased the opportunity costs of sending children to school. Such economic hardships were often noted in education circles. For example, the Hartog Committee (1929) had this to say on the subject of mass education. "In India, the great majority of parents who live on the land are poor, and their poverty is aggravated by improvidence and debt. Being illiterate and having an outlook confined almost entirely to their own surroundings and the daily routine of life, much persuasion is needed to convince them of the advantage of sending their children to school and keeping them there long enough to receive effective education, however, rudimentary. Even if schooling is free or school fees are small, the temptation to take a child away from school as soon as he is old enough to mind cattle or goats (which in an unfenced country has to be done by somebody) is great (p. 37)."

The opportunity cost of sending a child to school may have been high in colonial India, but was it higher than other predominantly rural economies? As shown earlier, primary school enrollment in India was among the lowest in the world even relative to countries at comparable levels of development that presumably faced similar opportunity costs. Moreover, the low level of education would suggest the returns to any schooling should be high because of the scarcity of skilled labor. Many of the statements on the perceived low returns to education are based on qualitative evidence, but what is needed is a quantitative assessment of the returns to education. Unfortunately, there are no such estimates for the colonial period. To accurately estimate returns, one needs individual or group information on wages by education and age. The official Price and Wage series only offer wages for skilled (common mason, carpenter and blacksmith) and unskilled workers, which are not necessarily the same as literate and illiterate workers. However, the official sources do reference wages for postmen and teachers, occupations requiring basic literacy skills.

We can use this information to generate crude estimates of the rate of return to literacy. Table 4 summarizes the exercise for a sample year, 1900. To assess returns in the absence of individual level data, we begin with wages in educated occupations such as primary school teachers and postmen. A few caveats are in order. First, these series underestimate teacher salaries because teachers received gifts in kind that are not included (Quinquennial Review 1901-02).¹⁶ Second, in most provinces the primary school teacher also served as the village

¹⁶The wages are reported as the average monthly salary of primary school teachers in schools for boys and girls. The wages are presumably also averaged over the different types of schools (public, aided and unaided), but the sources are unclear whether this is the case or not.

postman.

The annual rate of return is calculated as the difference between the wages of educated workers and unskilled or skilled workers divided by the annual cost of education (fees plus opportunity cost of time) multiplied by the number of years in school, which is 10 by assumption. In table 4, returns (1) calculates the rate of return of a primary school teacher relative to an unskilled agricultural worker assuming his opportunity cost is the same as his unskilled wage, he is in school for 10 years and attends a low fee primary school. Returns (2) adds the wages of postmen to those of primary school teachers and uses high fees for primary school. Returns (3) and (4) are similar to returns (2) except they are calculated relative to skilled wages and assume the opportunity cost is half the skilled wage when the individual is in school. Returns (3) uses low fees and returns (4) uses high fees.

Admittedly, the assumptions underlying these calculations are a bit ad-hoc, but they are designed to increase the costs of attending school. Thus, they should negatively bias the estimates. Regardless of the type of return, these estimates suggest very high returns to literacy averaging 33% per year for primary school teachers over unskilled agricultural workers and 20% per year for primary school teachers working as teachers and postmen over skilled workers. Teachers in Bombay and Central Provinces were very well paid, which is reflected in their relatively high returns to education. In comparison, returns are lower in Bengal because the teacher's salary included gifts in kind, a common practice in this region, that are excluded from the calculation.

This simple exercise misses several factors such as the probability of employment, wages in other educated occupations and discount rates to name a few. That said, they are similar to estimates of returns calculated for the post-independence period using more consistent data series. For example, Gounden (1967) estimates returns on the order of 16% to becoming literate. The returns to primary schooling average 20% across several studies from the 1960's and 70's. While the estimates in table 4 are higher, they do suggest returns to education were non-trivial. More research of this nature is needed to draw precise conclusions about the exact rate of return in the colonial period.

Low returns may be unable to account for India's poor performance in the primary sector, but there was low demand for basic education in many segments of the population. What then was driving this low demand? I believe this is related to several market failures in the education sector. Due to the high level of illiteracy in the population, parents were probably unaware of the specific costs and benefits of schooling. Even if they were aware of the benefits, poor people may have been unable to borrow money to finance their children's education because of weak capital markets. In a related vein, intergenerational commitment problems may have worsened both the credit market and information failures. These are

just potential hypotheses and there may have been other market failures at work. Future research should focus on sorting out and disentangling the role of these different market failures.

The returns explanation also fails to explain the widely diverging performance between the primary and secondary sector. Private returns to secondary education may have been high in the 19th century because well paying jobs as GOI clerks and inspectors often required matriculation certificates. But, anecdotal evidence suggests private returns fell in the early 20th century as Government jobs became harder to find and graduate unemployment increased.¹⁷ Moreover, estimates of private and social returns for the decades following Indian independence suggest returns to primary education were higher than secondary and higher secondary education.¹⁸ Given India's low level of literacy and economic development, one would imagine private and social returns to primary education were probably even higher in the colonial period.¹⁹

However, differential returns do offer a compelling explanation for the regional variation within British India. School enrollment and literacy were higher in the coastal provinces because they offered more economic opportunities for literate workers. The private demand for education was probably related to the size of the Government bureaucracy in urban cities and the opportunity for non-traditional employment in law and state offices. In other work, I argue colonial policies accounted for a third of the variation in education expenditures across provinces suggesting returns were primarily driving the remaining two-thirds (Chaudhary 2010).

4.2 Colonial Policies

There are many dimensions to education policy such as the amount of spending, the choice of public versus private schools, questions of curriculum, teacher training, etc. This section focuses exclusively on spending and evaluates whether public spending was adequate to the needs of mass education. I hope future research will take up other issues related to education policy.

Although the British created a new system of education, public investments in education were very small. Table 5 reports total public expenditures on education from 1881 to 1941. Public spending accounted for over 50% of total spending on education increasing to 60%

¹⁷See Nurullah and Naik (1951) and Basu (1974) along with Quinquennial Reviews of the period.

¹⁸See Gounden (1967), Blaug (1980), Psacharopoulos (1973), Heyneman (1980) and Asaoka (2006) provide a summary of rates of return studies in India.

¹⁹Psacharopoulos (1973) has studied rates of return to education for a broad cross-section of countries and concludes that rates of return are generally higher in developing countries as compared to developed countries and returns are always highest for primary education.

in the 1930's as the GOI increased grants to expand mass schooling. Despite the absolute increase, education accounted for a small share of the total budget averaging 3.5% in these decades. As late as 1931 the GOI allocated less than 1 rupee per-person to education and public spending accounted for 0.74% of national income. In comparison, public investments on education increased to 1.2% of national income in 1950-51 and then doubled to 2.4% by 1960-61 (Gounden 1967).

These numbers are especially low when placed in an international context. Government expenditures on human capital in British India were among the lowest in the world from 1860 to 1912 (Davis and Huttenback, 1986).²⁰ Government expenditures per capita averaged less than 0.01 pounds in British India, lower than average government spending in the Indian Princely States (0.02), in underdeveloped countries like Brazil and Mexico (0.05), and in other dependent British colonies (0.18). Similar to the patterns on absolute spending, human capital investments were a smaller share of the government budget in British India (4.1%) compared to countries at comparable levels of development that perhaps faced similar rates of return to schooling. For example, the Indian Princely States allocated 10% of their budget to human capital and foreign underdeveloped countries averaged 6%.

Official reports suggest British administrators were aware that spending was perhaps inadequate to meet the needs of expanding basic education, but they were also very critical of more spending leading to better outcomes. Official rhetoric often emphasized low demand as the primary constraint on educational development. This position may reflect colonial strategy to absolve the GOI from any blame for the low level of investment. On the other hand, Congress leaders and Indian nationalists bemoaned the low public spending and advocated higher spending as the key to better outcomes. The two positions mirror contemporary debates on the causal effect of public spending on educational outcomes (Glewwe, 2002).

Chaudhary (2010) offers the first credible causal estimates of local public expenditures on literacy. Using a new district-level dataset on spending and literacy, I found that public investments on primary education had a positive and significant effect on male literacy. However, there were no similar effects on female literacy. Official British opinion and Indian opinion were both partially correct: higher public spending would have increased male literacy, but building more public schools was not the answer to the severe and persistent problem of female illiteracy.

²⁰The numbers in the following discussion are from Davis and Huttenback (1986), chapter 4. Appendix 4.1 in their book provides more details on the data. Education is the dominant category of human capital expenditures, which also include spending on medicine, charity, relief, immigration and on occasion religion (as a matter of convenience when it was difficult to separate religious spending from education). For a small sample of countries and colonies, I double-checked reported public education expenditures against the human capital expenditures reported in Davis and Huttenback (1986) and they appear to follow the same pattern.

While low public spending cannot explain the poor absolute or relative performance of Indian women, differential economic returns are not a convincing explanation either. Low female literacy combined with an ever pressing need for female teachers in the colonial period suggests returns to female education must have been significant. At the very least, it is hard to believe returns by gender were that far apart to account for the wide gender gap in educational attainment. The divergence between men and women in education also parallels the divergence in labor market performance; for example, women were replaced by men in higher paying industrial jobs over the colonial period. Whether differences in education contributed to differences in labor market performance, or vice-versa, is unclear. But, the problem of female illiteracy does appear to be policy invariant because even today female literacy lags behind male literacy (65% compared to 82% in 2011). Many studies have looked at gender differences in contemporary India, but much more work is needed to understand the historical roots of such differences, their long run persistence and the economic consequences for Indian society.

In the case of male literacy, the calculations suggest public spending would have increased literacy. But, the GOI may have been reluctant to increase spending because it was not cost effective. The estimates in Chaudhary(2010) suggest it would have cost roughly 3 rupees in 1911 to make an additional male literate. One way to assess whether the cost of 3 rupees is big or small is to consider whether a simple reallocation of the entire public education budget to primary education would have been sufficient. Primary schools received less than half of public education funds and thus the reallocation would not be insignificant.²¹ A redirection of these expenditures to primary education would have increased total literacy by only 2.4 percentage points in 1921 from, say, 7.5 to 9.9 percent.²² Thus, a simple reallocation of the budget was not enough. The GOI needed to substantially increase public spending to achieve any significant literacy gains.

Several education reports also made crude calculations of the potential costs of a large scale expansion of the primary sector. For example, the Hartog Committee in 1929 estimated a total cost of rupees 19.5 crores on a recurring basis to introduce compulsory education covering more than 80 percent of the school age population.²³ But, even as late as 1941

²¹Expenditures on primary schools averaged only 34.3% of public education expenditures from 1891/92 to 1916/17. The share was fairly stable over this period—it was 32% in 1891/92, went up to 42% in 1901/02 but came back down to 31% in 1911/12. See *Progress of Education in India: Quinquennial Review (1917-22)*, vol. II, page 125. In comparison, the United States allocated more than 90% of public education expenditures to primary education from 1850 to 1890 and devoted larger public resources to secondary schooling only in the early 1900s when over 85% of the population was able to read and write. The United Kingdom allocated 73% of public education funds to primary schooling in 1890, while Japan allocated 84% to primary schools. Calculations are based on Lindert (2004) Volume 2 (Appendix Tables C1 and C3).

²²See Chaudhary (2010) for discussion of this issue.

²³The Report estimated a cost of 8 rupees per male student. This is higher than my estimate of 3 rupees

public expenditures on education were 18 crores of which less than 50 percent were allocated to primary education. It would appear the schemes to expand mass education were never seriously implemented.

Thus, the most significant failure of colonial policy was inadequate public spending both in absolute levels and as a proportion of the colonial budget. These additional public funds would have been effective at increasing male literacy, but not female literacy. Future research should focus on identifying why the GOI was reluctant to increase public spending. Clearly, public money had to come from somewhere. Perhaps, the rate of return to educational investments by the GOI was lower than other public investments? Was the GOI hesitant to promote mass education because of the potential socio-economic implications for colonial rule? Or, did the cost-benefit calculation not work in the Government's favor because India lacked the necessary tax institutions to extract the surplus from an educated work force? Personal incomes were subject to taxation, but the institutional set up to either increase the rates or extend them to a larger share of the population was perhaps politically infeasible in the 20th century. These are just potential hypotheses and questions—a systematic economic investigation of the GOI's motives and constraints remains to be done.

4.3 Caste, Religion and Fragmented Societies

By underinvesting in public education, colonial rule did constrain the development of primary education in India. But, this does not imply India would have enjoyed better outcomes as an independent state. If anything, the slow progress after 1947 underscores that inadequate funding was not the only problem. The presence of numerous castes and religions combined with the hierarchical divisions in Indian society seriously undermined private and public attempts to expand basic education.

Indian elites, defined by caste, wealth and occupation, were among the chief beneficiaries of English education. But, many of these same elites actively blocked schemes for public expansion of primary schooling. Landed elites were reluctant to support public education because they had to bear a disproportionate cost in terms of land taxes, the main source of local revenues for public primary schools. Educated elites belonging to the new urban intelligentsia were unlikely to promote mass education because it would increase competition for the much sought after GOI jobs.

Such resistance frequently occurred at different levels of government, either through the actions of local boards where landed and educated elites were disproportionately represented, or through direct lobbying of colonial officials. For example, Mr. C. T. H. Johnson, a

per student, which is calculated for 1911 and controls for the fact that districts with higher private demand would bear part of the cost.

district officer in Madras province, told the committee working on The Report of the Royal Commission upon Decentralization in India in 1908 that, “The Local Boards represent the monied, educated and land-owning classes; they are not really in favor of increased primary education, because it makes labor more difficult to handle; they are not in favor of a reduction of lower secondary education because they like to have the lower secondary schools to which men of their type send their children.” A fair number of landed Indian elites joined the official opposition to Gokhale’s initial attempts to introduce compulsory education in 1911. Mandal (1975) and Mukhdopadhyay (1983) provide similar accounts of how the Bengali Hindu Bhadrals hurt the cause of mass education in that province.

While public schools in principle were open to all members of society, in practice it was very difficult for the lower castes and classes to attend these schools. The Quinquennial Review of 1901/02 references a case where “hasty action on the part of local officers of the Kaira District in requiring the admission of low caste pupils led to five or six large schools being closed for years, and to the huts and crops of the low-caste people being burnt in one village (p. 393).” Schools in Madras were often located either in temples or parts of the village where Panchamas were not allowed. On account of such intimidation, special schools were opened for the lower castes either by the GOI or by private missionaries. But, it was often difficult to find qualified teachers. Such school segregation also increased the public costs because villages could not exploit the economies of scale in school provision.

The effects of caste and religious heterogeneity were even worse in the private sector because the GOI had limited control over private schools even those that received public subsidies. Using district level data on schools and local conditions, Chaudhary (2009) finds that characteristics of the local elite strongly influenced the provision of primary schools. Brahmans and other educated upper castes successfully directed private and, to a smaller extent, public resources to secondary schools for their children. Districts with a greater share of Brahmans had more public and private secondary schools plus a smaller ratio of primary to secondary schools. Districts with high levels of caste and religious diversity had fewer private primary schools and a smaller ratio of primary to secondary schools. However, upper castes were unable to completely co-opt the public policy-making process because districts with larger proportions of lower castes and Muslims also had more public secondary schools. But, these official attempts to circumvent the local politics of school provision were not particularly effective at increasing literacy.

In addition to caste problems, a heavy reliance on religious schools hurt the progress of Muslim literacy. Muslims in heavily Muslim dominant districts had worse literacy because they had experienced a more recent collapse of Muslim political authority and hence had more powerful and better funded religious authorities. The religious authorities established

religious schools, which were less effective at promoting literacy on the margin than public schools (Chaudhary and Rubin, 2011).

Colonial policies did not do much to ameliorate these long standing inequalities between groups. Public spending was too low and susceptible to elite capture at many levels. Since official policy could not override the local opposition to members of the lower caste attending public schools with the rest of the population, separate schools were created for these groups. But, these public secondary schools were not as effective at increasing literacy as the primary schools (Chaudhary 2009). A more focused effort at providing primary schools may have been more successful.

5 Conclusion

This chapter reviewed the basic educational patterns in British India and placed the poor Indian performance in a comparative perspective. Over the early 20th century, enrollment rates made slow but steady progress. Unfortunately, literacy was stagnant for most of the period emphasizing the poor quality of many of the schools. Back of the envelope calculations suggest returns to literacy were not as low as suggested by qualitative accounts, rather, inadequate public spending and social divisions within Indian society jointly hurt the progress of basic education.

The disadvantages of heterogeneous societies for public goods provision are not unique to India. Many recent studies have found similar negative effects of heterogeneity in countries ranging from the United States to Kenya (Alesina, Baqir and Easterly 1999, Miguel and Gugery 2005). In the Indian case, the inequality between castes further heightened these social divisions. I hope future researchers will compare and contrast the Indian experience to other parts of the world with similar levels of heterogeneity and inequality.

India's poor schooling record also has important implications for her trajectory of growth and development. In an influential paper, Acemoglu, Johnson and Robinson (2001) argue that non-settler colonies such as India fell behind because of poor quality institutions, or more specifically extractive institutions set up by the colonizers. In this view, weak property rights are a key example of extractive institutions. On the other hand, Glaeser et. al (2004) argue for the primacy of human capital over institutions. By the early 20th century, the colonial Indian state was not particularly extractive: tax rates were low by international standards and property rights by most accounts were decent. But, human capital development was severely de-emphasized, almost ignored. It would appear then that education was an important constraint on economic growth. But, before we draw definitive conclusions, future research needs to better explore the links between human capital, the rate of public and

private investment, and technological adoption in the Indian context. I hope this chapter spurs research on the general relationship between human capital and growth in India, as well as the specific constraints on education development outlined throughout the chapter.

6 References

1. Acemoglu, Daron, Simon Johnson and James Robinson. "Colonial Origins of Comparative Development: An Empirical Investigation." *American Economic Review* 91, no. 5(2001): 1369-1401.
2. Adam, W. Third Report on The State of Education in Bengal Including Some Account of the State of Education in Bear, Calcutta, 1838.
3. Alesina, Alberto, Reza Baqir and William Easterly. "Public Goods and Ethnic Divisions", *Quarterly Journal of Economics* 114, no. 4 (1999): 1243-1284.
4. Banerjee, Abhijit, and Rohini Somanathan. "The Political Economy of Public Goods: Some Evidence from India" *Journal of Development Economics* 82, no. 2 (2007):287-314.
5. Basu, Aparna. *The Growth of Education and Political Development in India 1898–1920*. Delhi: Oxford University Press, 1974.
6. _____. *Essays in the History of Indian Education*. New Delhi: Concept, 1982.
7. Becker, Gary S. *Human Capital: A Theoretical and Empirical Analysis with Special Reference to Education*. New York: National Bureau of Economic Research, 1964.
8. Bellenoit, H. J. A. *Missionary education and empire in late colonial India, 1860-1920*. London: Pickering & Chatto, 2007.
9. Benhabib, Jess and Mark M. Spiegel. "The Role of Human Capital in Economic Development: Evidence from Aggregate Cross-Country Data." *Journal of Monetary Economics* 34, no. 2 (1994): 143-74.
10. Blaug, Mark, P R G. Layard, and Maureen Woodhall. *The Causes of Graduate Unemployment in India*. London: Allen Lane, 1969.
11. McCully, Bruce Tiebout. *English education and the origins of Indian nationalism*. Gloucester, Mass: P. Smith. 1966.

12. Chand, Gyan. Local Finance in India. Allahabad: Kitabistan, 1947.
13. Chaudhary, Latika. "Determinants of Primary Schooling in British India", *Journal of Economic History* 69, No. 1 (2009): 269-302.
14. _____. "Taxation and Educational Development: Evidence from British India", *Explorations in Economic History* 47, No. 3 (2010): 279-293.
15. _____. "Land Revenues, Schools and Literacy: A Historical Examination of Public and Private Funding of Education", *Indian Economic and Social History Review* 47, no. 2 (2010): 179-204.
16. Chaudhary, Latika and Jared Rubin. "Reading, Writing and Religion: Institutions and Human Capital Formation." *Journal of Comparative Economics* 39, no. 1 (2011):17-33.
17. Davis, Lance E. and Robert A. Huttenback. *Mammon and the Pursuit of Empire: The Political Economy of British Imperialism, 1860-1912*. New York: Cambridge University Press, 1986.
18. Drèze, Jean and Amartya Sen. *India: Economic Development and Social Opportunity*. Clarendon, Oxford, 1998.
19. Easterlin, Richard. "Why Isn't the Whole World Developed?" *Journal of Economic History*, 41 (1981):1-19.
20. Ghosh, Suresh C. *The History of Education in Modern India 1757-1998*. New Delhi: Orient Longman, 2000.
21. Glaeser, Edward L., Rafael La Porta, Florencio Lopez-De-Silanes and Andrei Shleifer. "Do Institutions Cause Growth?" *Journal of Economic Growth* 9 (2004):271-303.
22. Glewwe, Paul. "Schools and skills in developing countries: education policies and socioeconomic outcomes." *Journal of Economic Literature* 40, no. 2(2002): 436 - 82.
23. Gounden, A. M. Nalla. "Investment in Education in India." *Journal of Human Resources* 2, no. 3 (1967):347-58.
24. House of Commons. "Despatch to Government of India on General Education in India." *Sessional Papers, 1854 (393), XLVII*.155.
25. House of Commons. "Progress of Education in India, 1892-97 (Third Quinquennial Review)." *Sessional Papers, 1899 (C.9190), LXV*.501.

26. _____. "Review of Progress of Education in India, 1897-98 to 1901-02. Fourth Quinquennial Review (East India: Education)." Sessional Papers, 1904 (Cd. 2181, 2182), LXV.1
27. _____. "Progress of Education in India, 1902-1907. Fifth Quinquennial Review (East India: Education)." Sessional Papers, 1909 (Cd. 4635, 4636), LXIII.1
28. _____. "Progress of Education in India, 1907-12. Sixth Quinquennial Review (East India: Education)." Sessional Papers, 1914 (Cd. 7485, 7486), LXII.1
29. _____. "Seventh Quinquennial Review, 1912-1917 (East India: Education)." Sessional Papers, 1919 (Cmd. 256, 257), XXVIII.1
30. Heyneman, Stephen P. "Investment in Indian Education: Uneconomic?" *World Development* 8, no. 2 (1980):145-63.
31. India. Report of the Indian Education Commission appointed by the resolution of the Government of India dated 3rd February 1882. Calcutta, 1883.
32. India. Census of India, 1901. Calcutta: Office of the Superintendent of Government Printing, India, 1903.
33. _____. Census of India, 1911. Calcutta: Office of the Superintendent of Government Printing, India, 1913.
34. Lindert, Peter. *Growing Public: Social Spending and Economic Growth since the Eighteenth Century*. Cambridge: Cambridge University Press, 2004.
35. Miguel, Ted and Mary Kay Gugerty. "Ethnic Diversity, Social Sanctions and Public Goods in Kenya." *Journal of Public Economics* 89, no. 11-12 (2005):2325-68.
36. Mincer, Jacob. *Schooling, Experience and Earnings*. New York: Columbia University Press, 1974.
37. Mukhopadhyay, Gopal. *Mass Education in Bengal (1882-1914)*. Calcutta: National Publishers, 1984.
38. Nurullah, Syed and J.P. Naik. *A History of Education in India (During the British Period)*. London: Macmillan, 1951.
39. Psacharopoulos, George. *Returns to Education: An International Comparison*. Amsterdam: Elsevier, 1973.

40. Richey, James A. Progress of education in India, 1917-1922. Calcutta: Supt. of Govt. Press, 1924.
41. Schultz, Theodore W. "Reflections on Investment in Man." Journal of Political Economy 70, no. 5 (1962):1-8.
42. _____. Investment in Human Capital. New York: Free Press and Macmillan, 1971.
43. _____. Transforming Traditional Agriculture. Chicago: University of Chicago Press, 1983.
44. Srinivas, M.N. Village, Caste, Gender and Method: Essays in Indian Social Anthropology. New York: Oxford University Press, 1996.
45. Whitehead, Clive. "The Historiography of British Imperial Education Policy, Part 1: India." History of Education 34, no. 3 (2005):315-29.

Figure 1: Total Enrollment, 1850-1940 (as a % of the population)

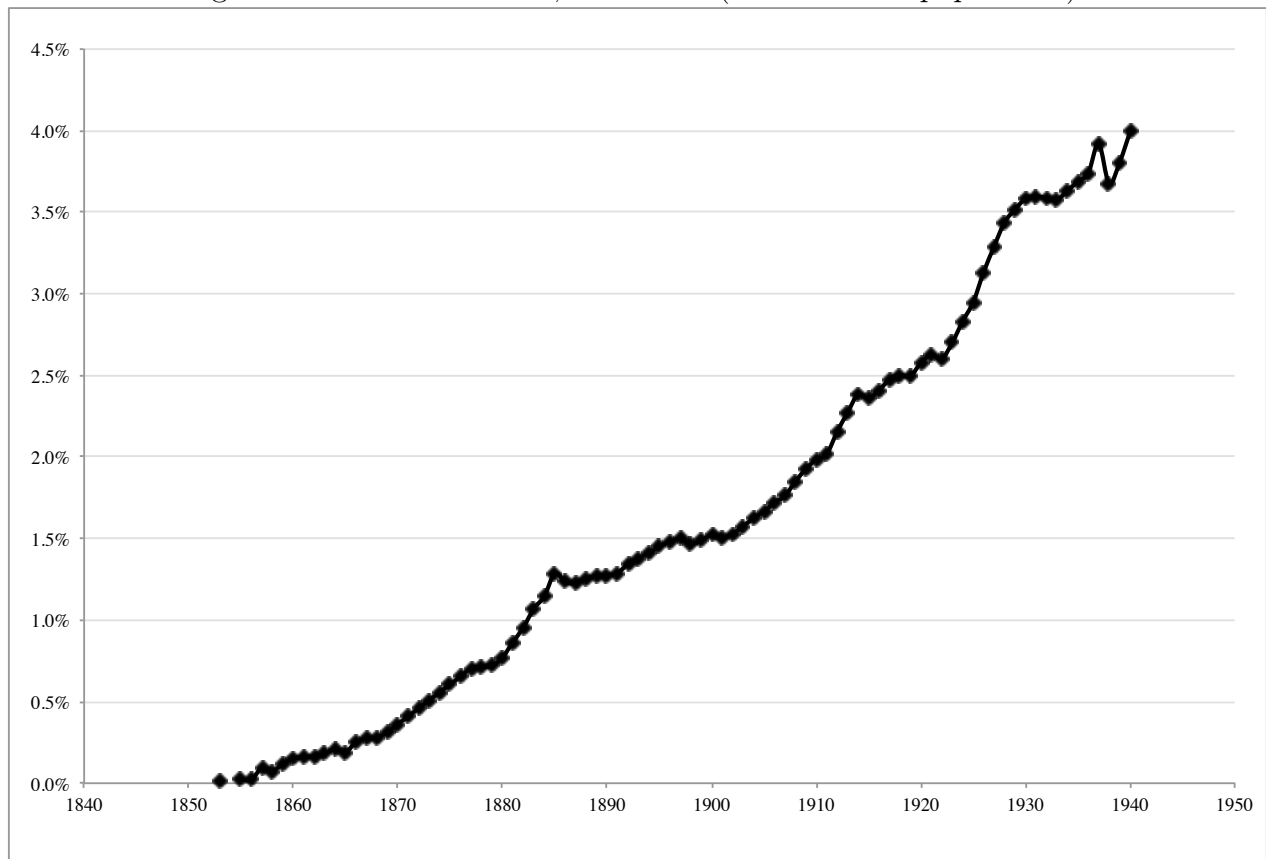


TABLE 1: Enrollment Rate (Pupils/School-Age Population)

	1891-92	1901-02	1911-12	1921-22	1931-32
<u>All Recognized Institutions:</u>					
BRITISH INDIA	9.6%	10.8%	16.0%	20.9%	30.2%
<i>Bengal</i>	12.7%	13.5%	24.4%	26.2%	36.2%
<i>Bihar and Orissa</i>			15.5%	15.0%	18.4%
<i>Bombay (inc. Sind)</i>	14.0%	14.9%	20.7%	30.9%	39.7%
<i>Central Provinces & Berar</i>	7.0%	8.9%	13.4%	15.8%	19.4%
<i>Madras</i>	11.8%	12.9%	18.6%	27.5%	41.0%
<i>Punjab</i>	4.8%	6.0%	10.6%	17.8%	33.9%
<i>United Provinces</i>	3.1%	5.2%	8.8%	14.2%	20.1%
<u>Primary Schools:</u>					
BRITISH INDIA	8.1%	8.9%	13.0%	17.0%	23.6%
<i>Bengal</i>	10.7%	11.0%	17.7%	20.5%	28.2%
<i>Bihar and Orissa</i>			13.0%	13.5%	15.6%
<i>Bombay (inc. Sind)</i>	12.8%	13.4%	18.6%	27.5%	34.9%
<i>Central Provinces & Berar</i>	5.7%	7.4%	10.9%	12.5%	14.3%
<i>Madras</i>	10.3%	10.9%	16.6%	24.4%	37.4%
<i>Punjab</i>	3.2%	3.7%	7.0%	10.3%	13.6%
<i>United Provinces</i>	2.2%	4.1%	7.2%	12.2%	16.5%
<u>Secondary Schools:</u>					
BRITISH INDIA	1.4%	1.7%	2.4%	3.3%	5.7%
<i>Bengal</i>	1.8%	2.2%	5.1%	4.7%	6.0%
<i>Bihar and Orissa</i>			1.2%	1.2%	2.3%
<i>Bombay (inc. Sind)</i>	1.1%	1.3%	1.8%	2.7%	3.8%
<i>Central Provinces & Berar</i>	1.2%	1.4%	2.4%	3.2%	4.8%
<i>Madras</i>	1.4%	1.8%	1.8%	2.7%	2.9%
<i>Punjab</i>	1.6%	2.1%	3.3%	7.2%	19.2%
<i>United Provinces</i>	0.8%	1.0%	1.4%	1.6%	3.0%
<u>Arts and Professional Colleges:</u>					
BRITISH INDIA	0.05%	0.06%	0.09%	0.16%	0.23%
<i>Bengal</i>	0.06%	0.09%	0.23%	0.31%	0.33%
<i>Bihar and Orissa</i>			0.04%	0.05%	0.08%
<i>Bombay (inc. Sind)</i>	0.05%	0.08%	0.12%	0.26%	0.38%
<i>Central Provinces & Berar</i>	0.01%	0.01%	0.03%	0.05%	0.10%
<i>Madras</i>	0.08%	0.08%	0.09%	0.17%	0.21%
<i>Punjab</i>	0.02%	0.06%	0.12%	0.20%	0.43%
<i>United Provinces</i>	0.04%	0.04%	0.08%	0.10%	0.16%

*Source: Progress of Education in India, Quinquennial Reviews (1891-1947). Data are not shown individually for the smaller provinces and . Upto 1911/12, the data include some native states of Bombay, Bengal, Central Provinces, & United Provinces but they represent 5% of the cc population. Secondary schools include high schools, middle English and middle vernacular schools. However, primary schools include middl schools for Bombay (all years) and for Madras from 1911-12. Bihar and Orissa are included in Bengal from 1891-1901. Enrollment and popu Bengal, Bihar and Orissa are from the Statistical Abstracts for 1911/12. School-Age Population is defined as 15% of the total population.

TABLE 2: Comparative Enrollment Rates (Number enrolled per 1,000 of the school-age population)

	India*	India-Lindert	Mexico	Brazil	Chile	Japan	France	Prussia	UK	USA
Primary School										
1880	.	42	187	70	111	306	816	711	549	906
1890	49	44	181	69	192	370	832	742	646	971
1900	53	47	185	102	245	507	859	732	720	939
1910	78	65	186	123	431	599	857	720	729	975
1920	102	80	231	147	422	602	704	758	701	924
1930	142	113	374	215	556	609	803	699	745	921
Secondary School										
1880	.	4	.	.	5	2	12	31	3	9
1890	8	8	1	.	3	3	11	32	3	20
1900	10	9	.	0	7	13	11	40	7	37
1910	14	12	2	5	12	74	14	44	21	55
1920	20	16	.	6	56	108	16/24	92	44	110
1930	34	28	5	8	52	165	19/32	94	58	193

Source: The India series in column 1 uses 25% of the total population to define the population aged 5 to 14 similar to the population age group used by Lindert. The Lindert series are drawn from Lindert, *Growing Public* (2004), vol. 2 (appendix tables C1 and C3).

TABLE 3: Literacy (in percentages)

	Ages 10 & over								English Literacy 1931	
	MALE				FEMALE				M	F
	1901	1911	1921	1931	1901	1911	1921	1931		
BRITISH INDIA	12.9	14.0	16.1	17.8	0.9	1.4	2.2	3.0	2.24	0.29
Assam	8.9	11.7	14.4	16.9	0.6	0.8	1.5	2.3	2.20	0.15
Bengal	13.8	18.7	21.0	20.1	0.7	1.5	2.3	3.4	4.27	0.48
Bihar & Orissa		10.4	11.4	10.8		0.5	0.7	0.8	0.92	0.07
Bombay	14.8	15.8	18.1	18.6	1.1	1.7	3.0	3.0	2.86	0.56
CP & Berar	7.9	8.7	10.3	12.2	0.3	0.4	1.0	1.1	1.11	0.13
Madras	16.0	18.3	19.9	21.1	1.2	1.7	2.6	3.2	2.57	0.35
Punjab	8.6	8.4	9.0	11.2	0.4	0.8	1.1	1.8	1.88	0.19
United Provinces	7.5	7.8	8.5	10.4	0.3	0.6	0.8	1.2	1.09	0.13
1931 Literacy										
	All	Hindu	Muslim	Christian	Jain	Tribal	Brahman	Depressed Castes	Highest Literacy	Caste w/ Highest Literacy
BRITISH INDIA	8.3	8.4	6.4	27.9	35.3	0.7	33.4	1.6		
Assam	9.3	10.6	6.6	21.1	38.5	1.8	.	3.1	25.6	Ahom
Bengal	11.1	15.7	6.7	43.8	42.4	0.9	43.1	5.0	62.7	Baidya
Bihar & Orissa	5.3	5.5	5.6	12.0	38.3	0.7	19.2	0.6	36.2	Kayastha
Bombay	10.8	10.2	6.9	37.1	32.4	0.5	51.0	2.8	51.0	Brahman
CP & Berar	6.6	6.4	15.0	56.4	36.3	0.7	35.2	1.5	35.2	Brahman, Baniya
Madras	10.8	10.4	12.5	21.3	35.5	0.3	54.3	1.5	54.3	Brahman
Punjab	6.3	9.6	3.3	9.8	29.5	0.0	15.1	0.8	27.6	Khatri
United Provinces	5.5	5.0	5.7	28.4	35.9	0.0	15.9	0.5	44.65	Kayastha

Source: Census of India 1931, Volume 1, Part 1 - Report, Chapter IX, Subsidiary Tables I, II, III, and V. Individual data are not presented for the smaller provinces. Madras includes Cochin and Travancore States for Brahman. The all India Brahman literacy is just the average across provinces and princely states included in this table.

TABLE 4: Returns to Education?

Province	<u>Annual Wages in Rupees (1900)</u>		<u>Annual Fees Primary</u>				<u>Returns</u>			
	Unskilled	Skilled	Primary School Teacher	Postmen	Low	High	(1) PT/Unskilled Low Fees	(2) PT+Postman/ Unskilled High Fees	(3) PT+Postmen/S killed Low Fees	(4) PT+Postmen/S killed High Fees
Bengal	71	149	101	126	0.59	1.43	4%	21%	10.36%	10.35%
Punjab	87	231	306	117	0.59	1.43	25%	38%	16.61%	16.60%
Bombay	81	242	564	134	0.59	1.43	59%	75%	37.61%	37.58%
Central Provinces	50	145	294	115	0.59	1.43	48%	70%	36.25%	36.21%
Madras	51	225	183	120	0.59	1.43	26%	48%	6.93%	6.92%
UP	47	195	222	108	0.59	1.43	37%	59%	13.74%	13.73%

Source: Wages are from the Statistical Abstracts. Primary school teacher salary and school fees are from the Quinquennial Review (1901-02)

TABLE 5: Public Expenditures on Education

	Public Expenditures	Real Public Expenditures (1881 Rupees)	Real Public Expenditures per capita	As a % of		
				Total Education Spending	Total GOI Budget	National Income
1881-82	10,602,683	10,602,683	0.05	56.18%	1.50%	.
1891-92	15,618,184	11,595,318	0.05	51.17%	1.76%	.
1901-02	17,703,968	13,554,601	0.06	44.13%	1.68%	0.14%
1911-12	40,523,072	22,823,339	0.09	51.56%	3.47%	0.21%
1921-22	114,961,178	42,354,118	0.17	62.56%	5.18%	0.33%
1931-32	168,419,016	110,772,239	0.41	61.95%	7.65%	0.74%
1941-42	180,484,512	99,367,877	0.34	58.49%		0.52%

Source: Quinquennial Reviews for public expenditures, Statistical Abstracts for total GOI revenues and Sivasubramonian (1997) for national income.